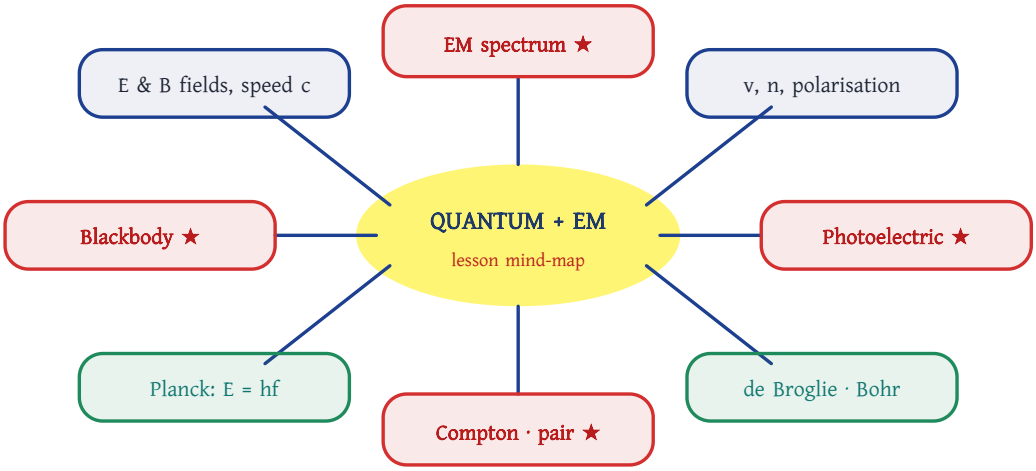
Quantum Physics & EM Waves

Spectrum · Planck · photoelectric · Compton · Bohr: exam-ready in 11 sheets



↑ the whole lesson on one page: red = high-yield, green = formula-driven

Syllabus coverage

"Electromagnetic waves, **Photoelectric effect**, **Quantum theory of light**, X-ray and X-ray diffraction, **Compton effect**". This notebook covers EM waves and the whole spectrum, propagation in media, **polarisation**, black-body radiation and Planck's quantum, the photoelectric effect, **Compton effect and pair production**, matter waves and **wave-particle duality**, and the Bohr model with the hydrogen spectrum. X-ray production and Bragg's diffraction law sit in the Basic Physics notes.

High-yield list

① $c = \lambda\nu$, no medium needed
② spectrum order **RMIVUXG**
③ $v = c/\sqrt{(\mu_r\varepsilon_r)}$
④ $K_{max} = hf - \varphi$
⑤ $\Delta\lambda = 2.43\ \text{pm}\ (1 - \cos\varphi)$
⑥ $\lambda = h/p$ · $E_n = -13.6/n^2$ eV

Legend: **yellow** = key term · **red circle** = must-memorise fact · **purple box** = memory trick · **red box** = exam trap · **green** = worked example

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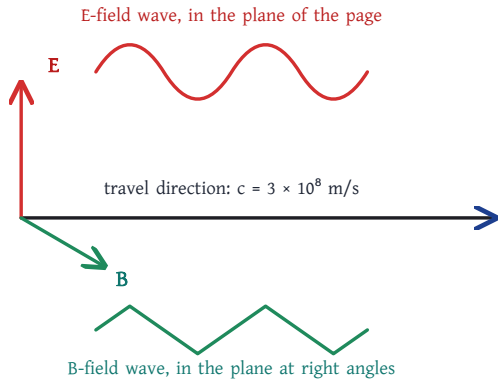
✓ EM waves: E, B and travel ..... p. 2
✓ The EM spectrum in order ..... p. 3
✓ Speed in a medium, n, polarisation ... p. 4
✓ Black-body radiation & Planck ... p. 5
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1 Electromagnetic Waves: Two Fields, One Wave

Definition: An EM wave is a travelling **coupled oscillation of an electric field E and a magnetic field B**. The two fields are **perpendicular to each other** and **both perpendicular to the direction of travel**. Because a changing E field creates a B field and vice versa, the wave needs **no material medium**.

Three lines to memorise: $c = 1/\sqrt{(\mu_0\varepsilon_0)} = 3 \times 10^8$ **m/s** · $c = f\lambda$ · in vacuum $E_0/B_0 = c$, and E and B are **in phase** (peaks together)

> E and B reach their peaks at the **same instant** — that is what "in phase" means.



E, B and travel are mutually perpendicular; E and B peak together

- **Origin:** Maxwell predicted EM waves (**1864–65**); Hertz generated and detected them (**1887**).
- **Nature:** transverse and uncharged, so an EM wave is **not deflected** by an electric or magnetic field — yet it still exerts forces on charges.
- **One speed for all:** every EM wave travels at c in vacuum; only f and λ change from band to band.
- **Energy flux** (Poynting vector) $S = E_0B_0/\mu_0$, directed along the propagation. Light itself is an EM wave (Maxwell, 1865).

Property	Electromagnetic wave	Sound wave
Medium needed?	No — travels through vacuum	Yes — gas, liquid or solid only
Nature	transverse	longitudinal
Speed in vacuum	3×10^8 m/s	cannot travel at all
Can be polarised?	Yes	No
What oscillates	electric **and** magnetic fields	pressure / particle displacement

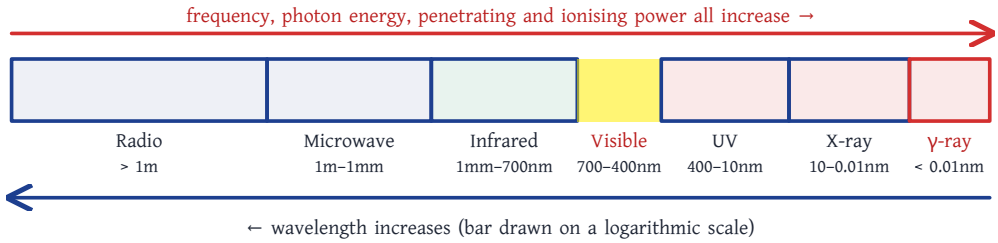
Memory trick: "**E–B–c** right": E, B and the travel direction are **mutually at right angles**, and the wave travels along **E × B**.

Exam trap: "EM waves are deflected by magnetic fields" is **false** — they carry no charge. But they **do** exert forces on charges, so the two statements are not the same.

Worked example ✍️: A station transmits on 90 MHz: $\lambda = 3 \times 10^8 / 90 \times 10^6 =$ **3.33 m** → a **radio** wave; on 900 MHz, $\lambda =$ **0.33 m** → a **microwave**. Photon energy there: 3.7×10^{-7} **eV**.

2 The Electromagnetic Spectrum

Definition: The **family of all EM waves**, arranged in order of wavelength. As λ falls, the frequency f, the photon energy hf, and the penetrating and ionising powers **all rise together**.



order of **decreasing** wavelength: radio → microwave → infrared → visible → UV → X-ray → gamma

Band	λ	f (Hz)	Photon E	Use
Radio	> 1 m	$< 3 \times 10^{8}$	$< 10^{-5}$ eV	broadcast, mobile
Microwave	1 m – 1 mm	$3 \times 10^{8} - 3 \times 10^{11}$	$10^{-5} - 10^{-3}$ eV	radar, Wi-Fi, oven
Infrared	1 mm – 700 nm	$3 \times 10^{11} - 4.3 \times 10^{14}$	10^{-3} – 1.8 eV	remote, night vision
Visible	**700 – 400 nm**	$4.3 - 7.5 \times 10^{14}$	1.8 – 3.1 eV	human eye, photos
Ultraviolet	400 – 10 nm	$7.5 \times 10^{14} - 3 \times 10^{16}$	3.1 – 124 eV	sterilising, fluorescence
X-ray	10 – 0.01 nm	$3 \times 10^{16} - 3 \times 10^{19}$	0.1 – 124 keV	radiography, crystals
Gamma ray	< 0.01 nm	$> 3 \times 10^{19}$	> 124 keV	cancer therapy, PET

colours inside visible light: violet **400** · blue **450** · green **495** · yellow **570** · orange **620** · red **700 nm**

Memory trick: **RMIVUXG** — Radio, Microwave, Infrared, Visible, Ultraviolet, X-rays, Gamma. In this order frequency increases; read it backwards for increasing wavelength.

Exam trap: The band boundaries are **the same numbers**: 1 m, 1 mm, 700 nm, 400 nm, 10 nm, 0.01 nm. Forgetting them costs the mark. Only **UV, X-ray and gamma** ionise.

Worked examples ✍️: (i) λ = 3 cm → f = 1×10^{10} Hz = 10 GHz → **microwave**. (ii) f = 3×10^{18} Hz → λ = **0.1 nm** → **X-ray**; E = 1240/0.1 = **12.4 keV**. (iii) λ = 600 nm → f = 5×10^{14} Hz, **orange**, E = **2.07 eV**.

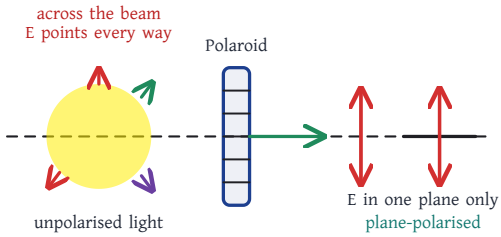
3 Speed in a Medium, Refractive Index, Polarisation

In a material medium: The wave drags the medium's charges along, so it slows down: $\mathbf{v = 1/\sqrt{(\mu\varepsilon)} = c/\sqrt{(\mu_r\varepsilon_r)}}$. The **refractive index** $n = c/v = \sqrt{(\mu_r\varepsilon_r)}$; for glass and water $\mu_r \approx 1$, so $n = \sqrt{\varepsilon_r}$.

λ changes in a medium. ν (frequency) **never** does — check every question for this.

Refraction:
Snell's law: $\sin i / \sin r = v_1/v_2 = n_2/n_1$ · entering a denser medium bends **towards** the normal, so r < i.

Photon energy hf never changes either, so the speed drops because λ shrinks: $\lambda' = \lambda/n$.



only **transverse** waves can be polarised, so every EM wave can

Polarisation in detail

Polarisation = restricting the vibration of E to one plane across the direction of travel. The **plane of polarisation** is the plane containing both the direction of travel and the electric vector. **Malus's law: $I = I_0 \cos^2\theta$** after a second filter at angle θ; unpolarised light through one ideal Polaroid drops to **half** the intensity. **Brewster's law: $\tan i_p = n_2/n_1$** — at that angle $i_p + r = 90°$, the reflected ray is **perpendicular** to the refracted ray, and the reflected light is completely plane-polarised with E perpendicular to the plane of incidence. Uses: sunglasses, camera filters, **LCD** panels, photoelastic stress analysis.

Medium	Vacuum	Air	Water	Crown glass	Diamond
Refractive index n	1	1.0003	1.33	1.50	2.42
ε_r if $\mu_r = 1$	1	≈ 1	1.77	2.25	5.86

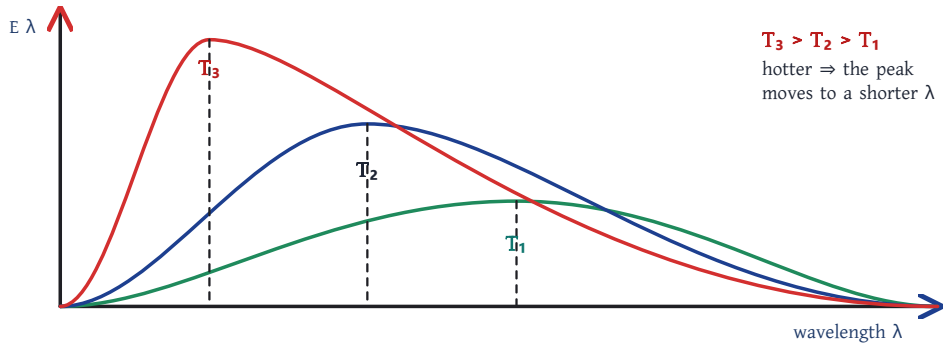
Memory trick: "**Enter, bend, stay**": light **enters** a denser medium, **bends** toward the normal, but its frequency **stays**. Only λ shrinks, λ′ = λ/n.

Exam trap: The commonest error is claiming the frequency changes when light enters glass or water. It does **not**. And sound cannot be polarised — it is longitudinal.

Worked example ✍️: 600 nm light in air enters glass of n = 1.5. (i) $f = c/\lambda = 5 \times 10^{14}$ **Hz** (unchanged). (ii) λ′ = 600/1.5 = **400 nm**. (iii) $v = c/1.5 = 2 \times 10^8$ **m/s**. (iv) A medium with $\mu_r = 4$, $\varepsilon_r = 9 \rightarrow n = \sqrt{36} = 6$, so $v = c/6 = 5 \times 10^7$ **m/s**.

4 Black-Body Radiation & Planck's Quantum

Definition: A **black body** is an ideal body that absorbs **all** EM radiation falling on it at every wavelength and reflects nothing. Kirchhoff's law: a good absorber is a good emitter, so at thermal equilibrium the emitted spectrum is the absorption spectrum.



every hot body radiates a black-body curve; hotter bodies are brighter everywhere and peak at a **shorter** wavelength

Law	Formula	Constant
Stefan–Boltzmann	$P/A = \sigma T^4$	$\sigma = 5.67 \times 10^{-8}$ W m^{-2} K^{-4}
Wien's displacement	$\lambda_{max} T = b$	$b = 2.9 \times 10^{-3}$ m·K
Rayleigh–Jeans	$e_\lambda \propto \lambda^{-4} T$ (long λ only)	fails at short λ
Planck's quantisation	$E = hf$	$h = 6.626 \times 10^{-34}$ J·s
Planck's radiation law	$e_\lambda d\lambda = 8\pi hc\, d\lambda / \lambda^5(e^{\{hc/\lambda kT\}} - 1)$	$k = 1.38 \times 10^{-23}$ J/K

Planck, 1900: Energy is exchanged in **discrete packets hf**, not continuously: a simple oscillator may only hold n·hf. That single assumption removed the **ultraviolet catastrophe**, in which classical theory demanded infinite energy at short wavelengths. As T rises, the peak slides IR → red → orange → yellow → white; the Sun's surface (≈ 5800 K) looks slightly bluish-white.

Memory trick: "**Wien finds the peak, Stefan counts the heat**": Wien ($\lambda_{max}T$ = const) gives T from one wavelength; Stefan–Boltzmann (σT^4) gives total power.

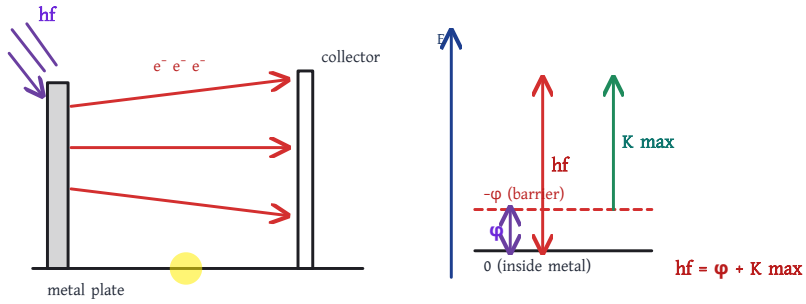
Exam trap: Wien's law finds the **peak wavelength**, not the total energy. Planck proposed quanta for **black-body radiation** in 1900; **Einstein (1905)** pushed them into light as photons.

Worked examples ✍️: (i) λ_{max} = 500 nm → T = $2.9 \times 10^{-3} / 5 \times 10^{-7}$ = **5800 K.** (ii) A room at 300 K → λ_{max} = **9.7 μm**, deep infrared. (iii) At 1000 K: P/A = $5.67 \times 10^{-8} \times 10^{12}$ = **5.67×10^4 W m^{-2}.** (iv) ν = 6×10^{14} Hz → E = hf = **3.98×10^{-19} J = 2.48 eV.**

5 The Photoelectric Effect

Definition: Emission of electrons from a metal surface when EM radiation of **frequency above the threshold** falls on it. Hertz observed it (1887), Lenard studied it (1902), **Einstein 1905, Nobel Prize 1921.**

Einstein's equation: $hf = \varphi + K_{max}$; φ = work function = $h f_0$; f_0 = threshold frequency; $\lambda_0 = hc/\varphi$; stopping potential: $eV_0 = K_{max}$.



left: the cell · right: one photon's **energy budget** — it pays the work function first, the rest becomes kinetic energy

You change...	Photocurrent	K_{max}, V_0
Intensity ↑, f fixed	increases	no change
f ↑, intensity fixed	≈ no change	increases linearly
λ ↑ but still < λ_0	no change	decreases
f < f_0, any intensity	no emission at all, however bright · emission is instantaneous	

Memory trick: "**Brightness counts, energy ranks**": intensity decides **how many** electrons escape; frequency decides **how fast** they escape.

Exam trap: Doubling the intensity does **not** double K_{max}. The K_{max} vs f graph is a straight line of **slope h** through (f_0, 0).

- **Why wave theory fails:** no emission below f_0 however intense the light; K_{max} depends on f, not intensity; emission is instantaneous.
- **Photon picture:** one photon of energy hf is absorbed by one electron, so $hf = \varphi + K_{max}$.

Worked examples ✍️: (i) φ = 2.0 eV, λ = 400 nm → E = 1240/400 = 3.1 eV → K_{max} = 3.1 − 2.0 = **1.1 eV** → V_0 = **1.1 V**. (ii) λ = 500 nm gives V_0 = 1.48 V → φ = 2.48 − 1.48 = **1.0 eV**, f_0 = **6×10^{14} Hz**, λ_0 = **500 nm**.

IT link: Photodiodes in optical-fibre receivers, CCD / CMOS image sensors and solar cells all detect **single photons** — the same $hf = \varphi + K_{max}$ relation, read out as a current.

6 Compton Effect, Pair Production, Photons

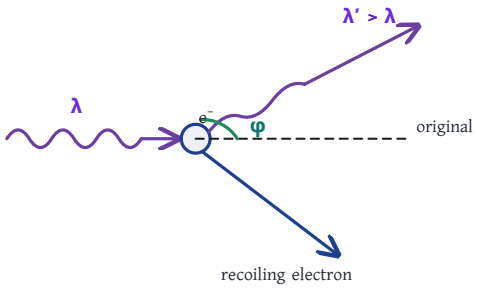
Photon: Light in packets: energy **$E = hf = hc/\lambda$**, momentum **$p = h/\lambda = E/c$**, **rest mass = 0**, charge 0, spin 0, speed always c. Because m = 0 while v = c, a photon's momentum and energy are always locked together by p = E/c.

Δλ is a property of the **electron** it hit, not of the photon: the same shift for any wavelength.

Compton effect: An X-ray photon scattered by a **nearly free electron** comes out with a **longer wavelength**. A.H. Compton (**1923**, Nobel **1927**) — direct proof that photons carry **momentum**.

Shift formula: $\Delta\lambda = \lambda' - \lambda = (h/m_e c)(1 - \cos\varphi)$, with the Compton wavelength $\lambda_C = h/m_e c$ = **2.43 pm**

φ	0°	60°	90°	180°
Δλ	0	1.21 pm	2.43 pm	4.85 pm (max)



the photon loses energy so λ grows; the electron recoils and takes the difference

The shift depends **only on φ** — not on the incident λ, not on the target material. The energy the photon loses becomes **kinetic energy of the recoiling electron**, so energy is still conserved.

Pair production & annihilation

- **Pair production:** a photon near a nucleus converts into an electron–positron pair, $\gamma \rightarrow e^- + e^+$. Energy must reach $2m_e c^2$ = **1.022 MeV**, so $\lambda \leq$ **1.21×10^{-3} nm**; the nucleus is needed to absorb the recoil momentum.
- **Annihilation:** $e^- + e^+ \rightarrow \gamma + \gamma$, and **each** photon carries at least **0.511 MeV**. Two photons, never one — a single photon would break momentum conservation.
- Photon count rises and energy per photon falls: mass is converted into radiation.

Memory trick: "0.511 **each**, 1.022 **together**": one electron's rest energy is 0.511 MeV; a photon must bring **twice** that to make a pair.

Exam trap: The shift is **independent of wavelength and material** — visible light would shift too, but by only ~2 pm out of 500 nm, which is undetectable. That is why the effect was found with X-rays.

Worked examples ✍️: (i) A 60 keV photon (λ = 0.0207 nm) scattered at 60°: Δλ = 2.43 × (1 − cos 60°) = **1.21 pm**, so λ′ = 0.0219 nm. (ii) Longest wavelength that can create a pair: λ = hc/E = 1240 / 1.022×10^6 = **1.21×10^{-3} nm**. (iii) At 180° the maximum shift is $2\lambda_C$ = **4.85 pm**.

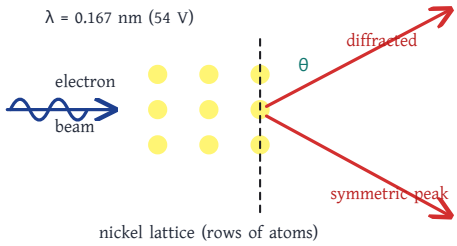
7 de Broglie Matter Waves & Wave-Particle Duality

Hypothesis, 1924: If light behaves like a particle, then **a particle must behave like a wave**. Louis de Broglie attached a wavelength to every moving particle: **$\lambda = h/p = h/mv = h/\sqrt{(2mK)}$**. Heavier or faster ⇒ shorter λ, so macroscopic objects have λ far too tiny to observe.

- **Davisson–Germer (1927):** electrons at **54 V** scattered from a **nickel** crystal gave a clear **diffraction pattern**. λ = 1.226/√54 = **0.167 nm** — the same order as the Ni lattice spacing, 0.215 nm — so matter waves were confirmed.
- **G. P. Thomson (1927)** sent electrons through a thin polycrystalline foil and got concentric **rings**, exactly like the optical diffraction rings of light through a small aperture.
- **Uses:** the **electron microscope** (short λ ⇒ tiny resolution), electron diffraction in a TEM, and LEED for reading chip surfaces.

Bohr orbits as standing waves

Only an **integer** number of de Broglie wavelengths fits round a stable orbit, **$2\pi r = n\lambda$**. Putting λ = h/p = h/mv gives **$mvr = nh/2\pi = n\hbar$**, Bohr's quantised angular momentum — now derived from wave physics. Phase velocity, group velocity, the wave function and the Schrödinger equation are taken further in the next lesson of this chapter; **$\lambda = h/p$** is all that is needed here.



a crystal is a **3-D diffraction grating** for electrons

Evidence for…	Observation	Experiment
Wave nature of light	interference, diffraction	Young's double slit (1801)
Particle nature of light	threshold f, instant emission	photoelectric effect
Particle nature of light	photon returns with a longer λ	Compton effect (1923)
Particle nature of light	$\gamma \rightarrow e^- + e^+$ above 1.022 MeV	pair production
Wave nature of matter	electron diffraction from a crystal	Davisson–Germer (1927)
Wave nature of matter	diffraction rings through a foil	G. P. Thomson (1927)

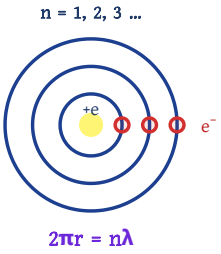
Memory trick: "**Light → particle** (photoelectric, Compton, pair). **Particle → wave** (de Broglie, Davisson–Germer). The **same h** sits in the middle of both halves.

Exam trap: Matter waves are not ripples of a medium. A photon has **zero** rest mass, an electron does not. Saying "light is a wave and a particle at once" is wrong.

Worked example ✍️: (i) 54 V: λ = 1.226/√54 = **0.167 nm**, comparable with the Ni spacing 0.215 nm, so diffraction is visible. (ii) A neutron of mass 1.675×10^{-27} kg at 1000 m/s has p = 1.675×10^{-24} kg·m/s, so λ = **3.96×10^{-10} m = 0.396 nm**.

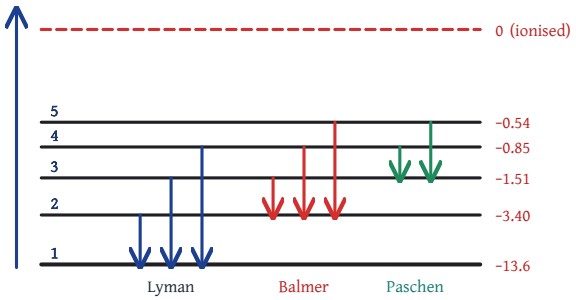
8 Bohr's Model & the Hydrogen Spectrum

Niels Bohr, 1913: (Nobel **1922**) ① electrons revolve only in certain **discrete stationary orbits** and radiate **no** energy while in them; ② angular momentum is quantised, $\mathbf{mvr = nh/2\pi = n\hbar}$; ③ a photon is emitted or absorbed only during a jump, with $\mathbf{hf = |E_2 - E_1|}$.



only whole wavelengths fit, so r is quantised

Quantity	Hydrogen (Z = 1)
Orbit radius	$r_n = n^2 \times 0.0529$ nm
Energy of level n	$\mathbf{E_n = -13.6 / n^2}$ **eV** ($E_1 = -13.6$ eV)
Ionisation from ground	13.6 eV
Rydberg formula	$1/\lambda = R(1/n_1^2 - 1/n_2^2)$
Constants	$R = \mathbf{1.097 \times 10^7\ m^{-1}}$, $a_0 = 0.529 \times 10^{-10}$ m



energy values in eV; a downward arrow emits a photon, an upward arrow absorbs one

Series	Jumps to	Wavelength range	Region	Famous lines
Lyman	n → 1	121.6 → 91.2 nm	ultraviolet	L_α = 121.6 nm
Balmer	**n → 2**	656 → 365 nm	**visible** (4 lines)	656.3, 486.1, 434.0, 410.2 nm
Paschen	n → 3	1875 → 820 nm	infrared	—
Brackett	n → 4	4051 → 2279 nm	infrared	—
Pfund	n → 5	7462 → 2279 nm	infrared	—

Memory trick: Only the series ending at **n = 2 (Balmer)** is visible; n = 1 gives **UV**, n ≥ 3 gives **IR**.

Exam trap: Bohr's model reproduces **hydrogen only** — it fails for helium and heavier atoms. E_n is **negative**; its magnitude shrinks as n grows.

Worked examples ✍️: (i) $E_2 = -13.6/4 = \mathbf{-3.4\ eV}$; ionising from n = 2 needs **3.4 eV**. (ii) n = 3 → 2: $\Delta E = 13.6(1/4 - 1/9) = 1.89$ eV → $\lambda = \mathbf{656\ nm}$, the red H_α line. (iii) Radius of n = 3: **0.476 nm**. (iv) Excitation n = 1 → 3 needs **12.09 eV**.

Night-Before-the-Exam Sheet

Constants: memorise these:

$c = 3 \times 10^8$ m/s

$h = 6.626 \times 10^{-34}$ J·s

$e = 1.602 \times 10^{-19}$ C

$m_e c^2 =$ **0.511 MeV**

$\sigma = 5.67 \times 10^{-8}$ W m^{-2} K^{-4}

hc = 1240 eV·nm

$\hbar = h/2\pi = 1.055 \times 10^{-34}$ J·s

$\lambda_C = h/m_e c =$ **2.43 pm**

$b = 2.9 \times 10^{-3}$ m·K (Wien)

$a_0 = 0.0529$ nm, $E_1 = -13.6$ eV

Formula sheet:

$c = 1/\sqrt{(\mu_0 \varepsilon_0)}$ · **$c = f\lambda$**

$\sin i / \sin r = n_2/n_1$

$P/A = \sigma T^4$ · $\lambda_{max} T = b$

$E = hf = hc/\lambda$ · $p = h/\lambda$

$f_0 = \varphi/h$ · $\lambda_0 = hc/\varphi$

pair: $2m_e c^2 = 1.022$ MeV

$\lambda = h/p = h/mv = h/\sqrt{(2mK)}$

$mvr = n\hbar$ · $2\pi r = n\lambda$

$1/\lambda = R(1/n_1^2 - 1/n_2^2)$

$v = c/\sqrt{(\mu_r \varepsilon_r)}$ · $n = c/v$

$I = I_0 \cos^2\theta$ · $\tan i_p = n_2/n_1$

$e_\lambda \propto \lambda^{-4} T$ (Rayleigh–Jeans)

$hf = \varphi + K_{max}$ · $eV_0 = K_{max}$

$\Delta\lambda = (h/m_e c)(1 - \cos \varphi)$

annihilation: ≥ 0.511 MeV each

e^-: $\lambda(\text{nm}) = 1.226/\sqrt{V}$

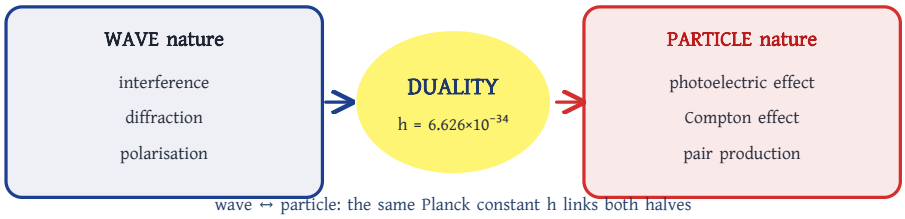
$E_n = -13.6/n^2$ eV · $r_n = n^2 a_0$

$\Delta E = 13.6(1/n_1^2 - 1/n_2^2)$ eV

Top 12 one-liners for MCQs

1. EM waves need no medium; sound does
2. $c = 3 \times 10^8$ m/s in vacuum only
3. EM waves are never deflected by E or B
4. Spectrum order: radio, microwave, IR, visible, UV, X, γ
5. Radio / microwave boundary = 1 m
6. Frequency never changes at a boundary
7. Only transverse waves polarise
8. Planck 1900: quantised black-body radiation
9. Einstein 1905: photoelectric effect
10. Brighter light ⇒ more electrons, not faster
11. Compton shift is independent of λ and material
12. Davisson–Germer: 54 V electrons on nickel

Top traps: f changes in a medium ✗ · brighter ≠ faster electrons ✗ · Planck ≠ Einstein for quanta ✗ · Bohr works for hydrogen only ✗ · only Balmer is visible ✗ · pair needs 1.022 MeV, annihilation 0.511 MeV each ✗



wave ↔ particle: the same Planck constant h links both halves

two halves, one constant

✓ Practice MCQs (exam pattern, 10 marks)

1. Speed of an EM wave in vacuum:
(a) 3×10^{6} m/s (b) 3×10^{8} m/s
(c) 3×10^{10} m/s (d) 9×10^{16} m/s

2. In an EM wave E and B are mutually:
(a) parallel (b) antiparallel
(c) perpendicular (d) identical

3. Energy of a 620 nm photon:
(a) 2.0 eV (b) 1.0 eV
(c) 4.0 eV (d) 0.5 eV

4. Photoemission needs a frequency f:
(a) below f_0 (b) equal to zero
(c) of any value (d) above f_0

5. Planck's constant h equals:
(a) 6.626×10^{34} J·s (b) 6.626×10^{-34} J·s
(c) 1.602×10^{-19} J (d) 3×10^{8} m/s

6. Maximum Compton shift, φ = 180°:
(a) 4.85 pm (b) 2.43 pm
(c) 4.85 nm (d) zero

7. Minimum photon energy for a pair:
(a) 0.511 MeV (b) 2.044 MeV
(c) 1.022 MeV (d) 8.19 MeV

8. Electron energy at n = 2 of H:
(a) −3.4 eV (b) −13.6 eV
(c) −6.8 eV (d) −1.51 eV

9. Visible hydrogen series:
(a) Lyman (b) Paschen
(c) Balmer (d) Brackett

10. Refractive index if $v = 2 \times 10^{8}$ m/s:
(a) 2 (b) 4
(c) 0.5 (d) 1.5

Answer key → 1-b · 2-c · 3-a · 4-d · 5-b · 6-a · 7-c · 8-a · 9-c · 10-d

Written question: State Einstein's photoelectric equation with every symbol explained. Light of wavelength 400 nm falls on a metal of work function 2.0 eV. Find (i) photon energy, (ii) K_{max}, (iii) stopping potential and (iv) threshold wavelength. Why can classical wave theory not explain the effect? 10 marks

1. **Equation:** $hf = \varphi + K_{max}$ — one photon's energy = work function + maximum kinetic energy of the ejected electron. 2
2. **(i)** $E = hc/\lambda = 1240 / 400 =$ **3.1 eV** **(ii)** $K_{max} = 3.1 - 2.0 =$ **1.1 eV** 3
3. **(iii)** $V_0 = K_{max}/e =$ **1.1 V** **(iv)** $\lambda_0 = hc/\varphi = 1240 / 2.0 =$ **620 nm** 3
4. **Why wave theory fails:** no emission below f_0 however intense the light; K_{max} depends on f, not intensity; emission is instantaneous. 2

✓ Know why EM waves need no medium
✓ Recite the spectrum order + boundaries
✓ Know f never changes at a boundary
✓ Convert λ ↔ f ↔ E with 1240/λ
✓ Solve $hf = \varphi + K_{max}$ in eV and volts
✓ Know Δλ and the pair thresholds
✓ Name the Balmer lines, Bohr radii

> **Before the exam ✓ · also practise (written):** ① sketch and label the EM spectrum · ② derive Snell's law from Huygens' principle · ③ explain Wien's displacement law